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Construction of chromosome-specific DNA libraries covering the whole genome of field bean (*Vicia faba* L.)

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Recombinant DNA libraries were constructed for seven chromosome types isolated from two translocation lines of field bean (*Vicia faba* L.) with reconstructed karyotypes. The chromosomes were selected so that the set of libraries covers the whole *V. faba* genome more than once. Individual chromosome types were highly purified by flow sorting, and their DNA was amplified by degenerate oligonucleotide-primed (DOP) polymerase chain reaction (PCR) and cloned into a plasmid vector. The choice of restriction site present in PCR primer and refinement of cloning protocol resulted in high cloning efficiency and allowed generation of libraries consisting of about 10^6 clones from 250 or 1000 sorted chromosomes. The insert size ranged between 50 and 2200 bp and the mean length estimated in individual libraries varied between 310 and 487 bp. Hybridization of cloned fragments with labelled genomic DNA showed that about 60% of inserts represented unique or low-copy sequences. The suitability of the libraries for genome mapping was demonstrated by isolation of clones containing microsatellite motifs.

Key words: chromosome flow sorting, degenerate oligonucleotide-primed polymerase chain reaction, chromosome-specific DNA libraries, genome mapping, *Vicia faba*

Introduction

High-density molecular maps are valuable tools for genome analysis, map-based gene cloning and marker-assisted breeding of crop plants. However, the large size and complexity of most of the higher plant genomes make the map saturation a difficult and tedious task. It is clear that division of the genome into smaller parts that can be studied separately would simplify and accelerate its analysis. This approach would also allow isolation of region-specific markers and integration of genetic and physical maps.

Chromosome separation by flow-sorting represents a powerful tool for the fractionation of eukaryotic genomes into well-defined parts represented by individual chromosome types (Gray & Cram 1990, Dolezel *et al.* 1994, 1995). The chromosomal DNA can be cloned into a suitable vector in order to generate a library that can serve as a source of molecular markers from the region of interest. This technique led to the preparation of chromosome-specific DNA libraries for each of the 24 human chromosomes. The libraries have been extensively used in genome mapping and in the search for markers linked to the genes involved in genetic diseases (Van Dilla & Deaven 1990). In this case, the libraries were prepared by direct cloning of purified DNA from 10^5–10^6 chromosomes into lambda and cosmid vectors. However, sorting of high numbers of chromosomes is time-consuming. Furthermore, long sorting times may affect the purity of sorted fractions negatively.

Recently, procedures based on PCR amplification have been developed that require only tens to hundreds of template chromosomes for library construction. One of the most widely used techniques, named degenerate oligonucleotide-primed polymerase chain reaction (DOP-PCR) was described by Telenius *et al.* (1992). The method allows general amplification of DNA sequences without a priori knowledge of their composition. It is based on multiple locus priming, which is achieved by employing oligonucleotides of partially degenerate sequence as primers and by non-stringent annealing conditions during initial PCR cycles. DOP-PCR using primer 6-MW has been shown to be suitable for amplification of genomic DNA in animal (Telenius *et al.* 1992, Langford *et al.* 1992, Breneman *et al.* 1993, Guan *et al.* 1994, Hoebee *et al.* 1994) as well as in plant species (Arumuganathan *et al.* 1994, Pich *et al.* 1994). Amplified fragments can subsequently be cloned via restriction site *Xho*I, which is present near the 5′ end of the primer (Telenius *et al.* 1992).

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Field bean (*Vicia faba* L., $2n = 12$) possesses one of the largest genomes among legume species (2C = 26.9 pg, i.e. 12 979 Mbp/haploid genome; Dolezel *et al.* 1992a, Bennett & Leitch 1995). In spite of recent progress in field bean linkage analysis (Torres *et al.* 1993), a detailed map is not available and most of the linkage groups remain unassigned to chromosomes. Recently, flow karyotyping and chromosome sorting has been established in this species (Lucretti *et al.* 1993, Dolezel & Lucretti 1995), and sorted field bean chromosomes have been shown to be suitable for physical gene mapping using PCR (Macas *et al.* 1993a). Here, we describe the construction of recombinant DNA libraries from two wild-type and five reconstructed chromosome types in *V. faba*. Chromosomes were separated by flow sorting, their DNA was DOP-PCR amplified, and the reaction products were cloned into a plasmid vector. To our knowledge, this is the first report on the construction of a complete set of chromosome-specific DNA libraries in plants.

Materials and methods

Plant material

Seeds of field bean (*Vicia faba* L., $2n = 12$) cv. 'Inovec' with a standard karyotype were obtained from Dr S. Ondro (Horná Streda, Slovakia). Seeds of field bean translocation lines EF and JF with reconstructed karyotypes were kindly provided by Dr I. Schubert (IPK Gatersleben, Germany). For a detailed description of chromosome translocations in these lines see Schubert *et al.* (1986).

Reagents

Enzymes, plasmid vector pBluescript II SK^+, bacterial strains and competent cells were supplied by Stratagene (La Jolla, CA, USA). Oligonucleotides were purchased either from TIB Molbiol (Berlin, Germany) or from the Department of Molecular Medicine, King's College School of Medicine and Dentistry (London, UK).

Chromosome isolation and sorting

Chromosome suspensions were prepared from synchronized root tips as described previously (Dolezel *et al.* 1992b) and stained with 4',6-diamidino-2-phenylindole (DAPI) at a final concentration of 0.5 μg/ml. Flow karyotyping and chromosome sorting were performed using a FACStarPlus flow cytometer and sorter (Becton Dickinson, San José, USA) according to Lucretti *et al.* (1993) and Dolezel & Lucretti (1995) with some modifications. An argon ion laser tuned to 351–360 nm with 100 mW output power and narrow beam elliptical optics were used for excitation. Fluorescence emitted from DAPI-stained chromosomes was collected through a 400-nm longpass filter. An autoclaved solution of 25 mM KCl was used as a sheath fluid. To ensure the highest purity of sorted fraction, approximately 50 000 chromosomes were sorted into 300 μl of LB01 buffer (Dolezel *et al.* 1989). From this enriched suspension, 250 or 1000 chromosomes were sorted into PCR tubes, each containing 60 μl of sterile water. Samples were transferred to −20°C immediately after sorting. To check the purity of sorted fractions, 100 chromosomes were sorted directly onto a microscopic slide before or after each sorting run. After drying, the slides were mounted in LB01 buffer containing DAPI and observed using an Olympus BX60 fluorescence microscope.

DOP-PCR amplification

The primers used for DOP-PCR reactions were either 6-MW (5'-CCG ACT CGA GNN NNN NAT GTG G-3', where n = A, G, C or T and the *Xho*I restriction site is underlined) or various modifications resulting in restriction sites other than *Xho*I: E6-MW (*Eco*RI site, GAATTC), P6-MW (*Pst*I site, CTGCAG) and K6-MW (*Kpn*I site, GGTACC). Reaction mix (100 μl) consisted of 1 μM primer, 0.2 mM dNTPs, 3 mM $MgCl_2$, 10 mM Tris-HCl (pH 8.8), 50 mM KCl, 0.01% (w/v) gelatin and 5 U of *Taq* DNA polymerase. PCR was performed as described by Telenius *et al.* (1992) with some modifications, using a Techne (Cambridge, UK) Cyclogene thermal cycler. The samples were denatured for 5 min at 95°C and then subjected to five cycles of 30 s at 95°C, 1.5 min at 30°C, 3 min transition to 72°C and 3 min at 72°C, followed by various numbers (25–35) of cycles of 30 s at 95°C, 1.5 min at 56°C and 3 min at 72°C, with the final extension step prolonged to 20 min.

Characterization of DOP-PCR products

Aliquots (10 μl) of amplification products were subjected to electrophoresis on 1% agarose gel and blotted onto Hybond N^+ membrane (Amersham, UK) using alkaline transfer. Hybridization with total *V. faba* DNA labelled by digoxigenin (DIG DNA Labelling Kit, Boehringer Mannheim, Germany) was performed at 68°C according to standard protocols (Ausbel *et al.* 1991). The final post-hybridization washes were done twice for 20 min in 0.1 × saline sodium citrate (SSC), 0.1% sodium dodecyl sulphate (SDS) at 68°C and the probe was detected using DIG Luminescent Detection Kit (Boehringer Mannheim).

To check the presence of low-copy genomic sequences, 1.5 - μl samples of purified DOP-PCR products were used in PCR with primer pairs specific for one of the following genes: vicilin (Vic-F: 5'-GCA GCT ACC ACA TTG AAA GAT TC-3', Vic-R: 5'-AAA TCG GCG TCA GTT TGT TG-3'; product length 281 bp), USP (USP-F: 5'-TCA ATC TTG GGC ACA CTA ATT CT-3', USP-R: 5'-TGG TCT TGA TAT GGA GCA AAG TT-3'; product length 256 bp) and legumin A (LegA-F: 5'-GGC TTC AAG AGG GAT TTC TTG GAA GAT GC-3', LegA-R: 5'-CCA TTG TCT CCA TGC CTT CTG CTT TCA CC-3'; product length 308 bp). The reactions were performed in a 30 μl volume of 0.2 μl primers, 0.2 mM dNTPs, 1.5 mM $MgCl_2$, 10 mM Tris-HCl (pH 8.8), 50 mM KCl, 0.01% (w/v) gelatin and 4 U/100 ml of *Taq* DNA polymerase, with initial denaturation at 95°C for 3 min, followed by 35 or 40 cycles of 30 s at 95°C, 50 s at 55°C and 1 min at 72°C, with the final extension for 6 min.

Cloning

The DOP-PCR products were extracted with chloroform and incubated with proteinase K according to Crowe *et al.* (1991) before purification. Samples were adjusted to 5 mM EDTA, 20 mM Tris-HCl (pH 8.0), 0.5% SDS and 50 μg/ml proteinase K. Following incubation for 30 min at 37°C, the enzyme was

inactivated by heating to 65°C for 10 min and by phenol–chloroform extraction. The DNA was then purified on QIAquick columns (Qiagen, USA) and eluted to 50 μl of 10 mM Tris-HCl, pH 8.5.

For evaluation of cloning efficiency, the PCR products generated by individual primers were cleaved by 40 U of the corresponding restriction enzymes for 4 h, heated to 65°C for 10 min, phenol extracted and again column purified. DNA concentration in individual samples was estimated by running aliquots on agarose gel electrophoresis and by their quantification using an Ultroscan XL densitometer (LKB Bromma, Austria). Equal amounts (400 ng) of the DNA were then ligated with 800 ng of pBluescript II SK^+, which was cleaved by corresponding enzyme, dephosphorylated and gel purified before use. The reaction was performed in a 20 μl volume of 1 × ligation mix (50 mM Tris-HCl, pH 7.5, 7 mM $MgCl_2$, 1 mM DTT, 1 mM rATP, 2 U of T4 DNA ligase) overnight at 12–16°C. The ligase was heat inactivated (65°C for 10 min) and 2 μl of the samples was used for heat-shock transformation of competent *E. coli* ('lab-made' competent *E. coli* XL-1 Blue MRF' or commercially supplied Epicurian Coli SURE and Epicurian Coli XL2-Blue MRF' cells). Aliquots of transformed bacterial suspensions were plated to determine the number of colonies.

The procedure for library production from flow-sorted chromosomes was essentially the same as described above. The primer K6-MW was used for DOP-PCR amplification of either DNA from sorted chromosomes or 2 ng of *V. faba* genomic DNA for a total of 35 cycles. One-fifth (10 μl) of the purified amplification product was ligated to *Kpn*I-digested and dephosphorylated plasmid. The ligation mix (2 or 3 μl) was then used for transformation of Epicurian Coli XL2-Blue MRF' ultracompetent cells. All transformed bacteria were plated onto LB medium containing 50 μg/ml carbenicillin and X-gal/IPTG. The bacteria from the individual plates of each library were then pooled together by resuspension in liquid LB medium, mixed with glycerol to its final concentration of 15%, divided into aliquots and stored at − 70°C for further use.

Characterization of DNA libraries

The proportion of colonies carrying non-recombinant plasmids was determined by blue/white assay on medium containing X-gal/IPTG and subsequent analysis of plasmids from selected colonies (about 50% of blue colonies also contained inserts). The insert size range was determined by PCR amplification of the plasmid polylinker region with T3 and T7 primers for at least 60 colonies from each library. Amplified inserts were dot-blotted onto Hybond N^+ membrane and hybridized with DIG-labelled field bean DNA to detect clones containing repetitive genomic sequences. The hybridization was performed overnight at 68°C in 5 × SSC, 2% (w/v) blocking reagent (Boehringer Mannheim), 0.1% N-lauroylsarcosin, 0.02% SDS and 100 μg/ml sonicated salmon sperm DNA. Then, the membranes were washed twice for 10 min in 2 × SSC, 0.1% SDS, for 10 min in 1 × SSC, 0.1% SDS, and twice for 20 min in 0.1 × SSC, 0.1% SDS at 68°C. The probe was detected by the DIG Luminescent Detection Kit.

Isolation and sequencing of clones containing microsatellites

Single colonies from the library Vf 1 JF01 were cultivated separately in 96-well microtitre plates. The clones containing $(GA)_n$ or $(AAC)_n$ sequences were retrieved by screening the pools from individual plates (and subsequently the pools of rows and columns of positive plates) by PCR using T3 or T7 primer in combination with the microsatellite primer $(GA)_{11}$ or $(AAC)_7$. Selected clones were sequenced by the dideoxy chain termination method (Sanger *et al.* 1977) using the Silver Sequence DNA Sequencing System (Promega, USA).

Results

Optimization of amplification and cloning procedures

The procedure for DNA amplification by DOP-PCR and subsequent cloning of reaction products was set up using *V. faba* genomic DNA as a template. The amounts of genomic DNA used in the reactions corresponded to the DNA content of 250–1000 chromosomes (0.1–10.0 ng). Individual steps of the protocol were optimized in order to achieve high cloning efficiency permitting production of complex libraries. In this respect, two key factors were determined: (1) the number of PCR cycles; and (2) the type of restriction site used for cloning. The DNA produced in DOP-PCR reactions after a large number of cycles or, generally with greater yields, tended to be difficult to clone. The samples amplified for a total of 40 cycles were also significantly more sensitive to mung bean nuclease treatment than those amplified for only 35 cycles (data not shown). These results indicated the presence of single-stranded DNA and suggested a limited capacity of the reaction to generate perfectly double-stranded fragments when the products were highly concentrated.

Since our preliminary experiments using the *Xho*I restriction site showed poor cloning efficiency, several modifications of the 6-MW primer with different restriction sites were tested. Four primers (6-MW, E6-MW, P6-MW and K6-MW, bearing recognition sites for restriction enzymes *Xho*I, *Eco*RI, *Pst*I and *Kpn*I respectively) were compared with respect to their ability to facilitate amplification of genomic DNA and its subsequent cloning. With the exception of E6-MW, all primers amplified *V. faba* genomic DNA, including low-copy genomic targets with high efficiency (data not shown). The influence on cloning efficiency of the type of restriction site present on the primer was obvious from the comparison made by ligating equal amounts of the cleaved DOP-PCR products generated by individual primers into the plasmid vector. Reproducible differences in the number of colonies were obtained in several experiments with the DNA generated in independent PCR runs and using either our competent or commercially available ultracompetent cells. The highest numbers of colonies were observed for K6-MW and E6-MW primers, slightly lower for P6-MW and usually 5–20 times lower for the primer 6-MW (typical results are given in Table 1). Based on these results, the primer K6-MW was chosen for construction of the libraries from flow-sorted chromosomes.

Table 1. Comparison of cloning efficiency of DOP-PCR products generated by individual primers

Primer	Number of colonies	
	SURE 2	XL-2 Blue MRF′
6-MW	38	2300
E6-MW	563	40600
P6-MW	288	11500
K6-MW	966	28300

Equal amounts of the reaction products were ligated to cleaved and dephosphorylated plasmid vector (see Materials and methods). Total numbers of colonies obtained after the transformation of two different *E. coli* strains by one-tenth of the ligation mixes are indicated.

Chromosome sorting

Since only one chromosome type can be discriminated and separated from the field bean wild-type karyotype by flow cytometry (Lucretti *et al.* 1993), we used EF and JF translocation lines with reconstructed karyotypes for chromosome isolation (Figure 1). As can be seen in Figure 2A, flow cytometric analysis clearly discriminated all six chromosome types in the EF line. However, sorted fractions of larger chromosomes (1 and 2) were contaminated with doublets of shorter chromosomes. In an analogous manner, fractions of smaller chromosomes were contaminated by chromatids of longer chromosomes. To increase the purity of sorted fractions, we have simultaneously analysed chromosome DNA content (fluorescence pulse area) and chromosome length (fluorescence pulse width). Sorting windows were set on bivariate flow karyotype as shown in Figure 2B. All chromosome types from this karyotype were sorted using a two-step sorting procedure to ensure the highest purity (see Materials and methods). Karyotype JF was used for the sorting of chromosome 1 JF. The degree of resolution of the flow karyotype was similar to that of EF and the peak of chromosome 1 was clearly discriminated (data not shown). Again, pure fractions were obtained after bivariate analysis and two-step sorting. Analysis of chromosome fractions sorted onto microscopic slides showed that contamination with other chromosomes and/or their fragments was generally about 5% (Table 3).

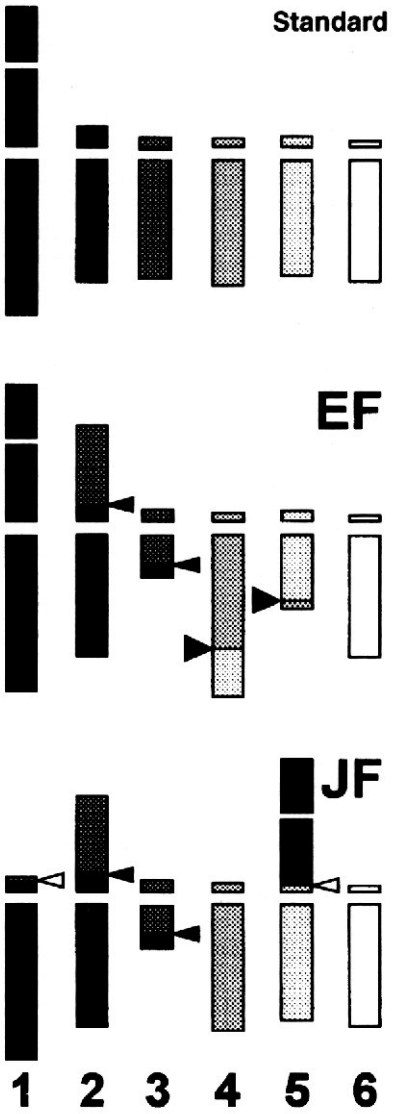


Figure 1. Idiograms of *V. faba* ($2n = 12$) lines with standard (wild-type) and reconstructed karyotypes (EF and JF) as used in this study. Arrows point to presumed translocation breakpoints (large arrow, translocation E; small arrow, translocation F; empty arrow, translocation J). Relative chromosome lengths were taken from Schubert *et al.*(1986).

Construction of libraries

DNA from sorted chromosomes was DOP-PCR amplified using primer K6-MW for a total of 35 cycles in 100 μl of reaction volume. The samples containing 2 ng of *V. faba* genomic DNA and samples with no DNA were run in parallel as controls. Analysis of the amplification products by agarose gel electrophoresis showed a smear of fragments (200–2500 bp) in all samples, except the control lacking DNA (Figure 3A). The gel was then blotted onto a nylon membrane and the reaction products were hybridized with DIG-labelled field bean DNA to confirm their origin (Figure 3B). In parallel, the aliquots of DOP-PCR products were used for reamplification with gene-specific primers to check their presence in individual amplified fractions. PCR reactions with primer pairs specific for vicilin, USP and legumin A sequences produced strong bands of the expected size in samples corresponding with the location of these genes determined in previous studies (Macas *et al.* 1993a,b) (Table 2 and Figure 4).

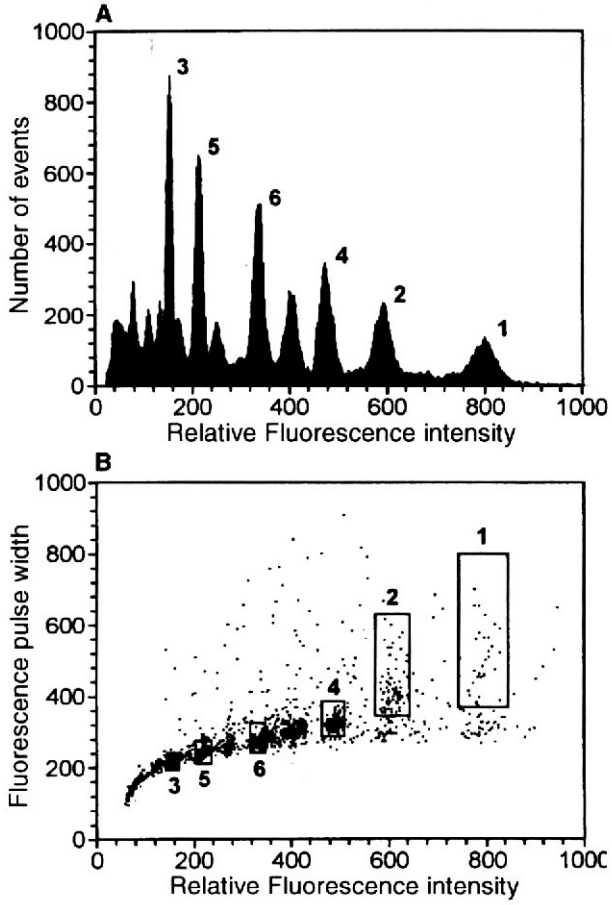


Figure 2. Flow karyotypes obtained after the analysis of DAPI-stained chromosome suspension prepared from *V. faba* translocation line EF. **A** Univariate analysis of DAPI fluorescence intensity. All six chromosome types are clearly resolved. **B** Bivariate flow karyotype: DAPI fluorescence intensity (fluorescence pulse area) versus fluorescence pulse width (corresponding to chromosome length). Rectangles indicate position of sorting windows for individual chromosome types.

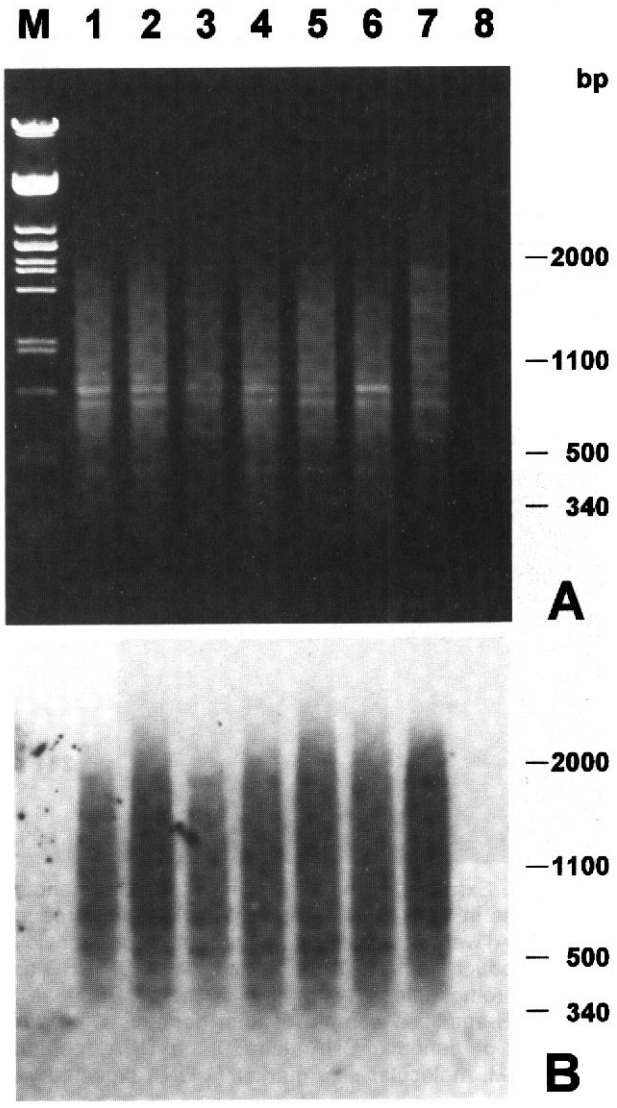


Figure 3. Analysis of DOP-PCR-amplified DNA from flow-sorted chromosomes. **A** Ethidium bromide-stained gel (1% agarose). M, molecular size marker (λ DNA digested with *PstI*); lanes 1–6, chromosomes 1–6 of the karyotype EF; lane 7, 2 ng of field bean genomic DNA; lane 8, no DNA (negative control). **B** Southern blot analysis of the amplification products probed with DIG-labelled field bean genomic DNA.

The largest part (80 μl) of the DOP-PCR products was then purified, digested by *Kpn*I and ligated to dephosphorylated pBluescript II SK$^+$ plasmid vector. Ligation mixes (2 or 3 μl) were used for transformation of competent *E. coli* cells. The libraries were constructed from all six chromosome types of karyotype EF, from chromosome 1 of karyotype JF and, as a control, also from 2 ng of total genomic DNA. Numbers of colonies obtained after transformation are given in Table 3.

Characterization of libraries

The total size of the libraries was estimated based on the results of transformation using aliquots of the ligation mixes. Considering the fact that only one-fifth of the amplified DNA was ligated and a small fraction of the ligation mix was used for transformation, the total number of clones available from a single sorted sample was estimated to be about 10^6 (Table 3). At least 60 randomly selected white colonies from each library were analysed for insert length by PCR amplification using T3 and T7 primers (Figure 5). The actual size was calculated by subtracting 175 nucleotides representing the amplified part of the plasmid polylinker from the fragment length determined by agarose gel electrophoresis. The size varied from 50 bp to 2200 bp with a predominance of inserts in the 100–800 bp range. The mean insert sizes for individual libraries are given in

Table 2. Detection of low-copy genomic sequences in amplified chromosomal DNA

Gene	Chromosome type							Control (genomic DNA)
	1 EF	2 EF	3 EF	4 EF	5 EF	6 EF	1 JF	
Vicilin	−	++	(+)	(+)	−	−	−	++
USP	++	+	−	−	−	−	++	++
Legumin A	−	−	−	+	−	−	−	(+)

Aliquots of DOP-PCR products were reamplified with primers specific for vicilin, USP and legumin A genes. The occurrence of PCR products of expected size is indicated as − (no product), (+), + and ++ (very strong band). Respective target sequences are located on chromosomes 2 EF (vicilin), 1 EF and 1 JF (USP) and 4 EF (legumin A) (Macas *et al.* 1993 a, b).

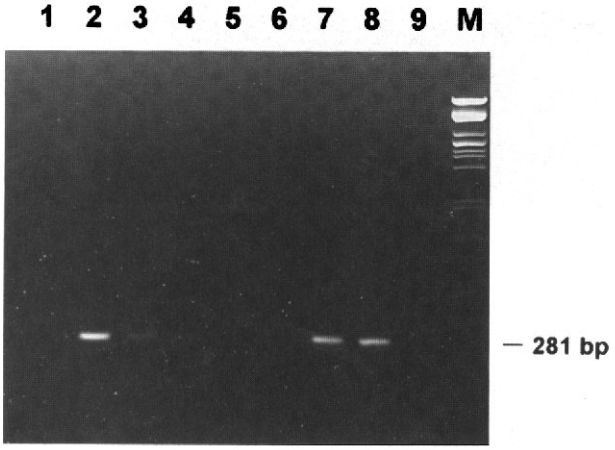


Figure 4. Example of the detection of low-copy genomic sequences in DOP-PCR products. Amplification products (Figure 3A) were reamplified with primers specific for vicilin gene (product length 281 bp). Lanes 1–6, DOP-PCR-amplified chromosomes 1–6 of the karyotype EF; lane 7, DOP-PCR-amplified sample of total genomic DNA; lane 8, 15 ng of genomic DNA; lane 9, no DNA (negative control). M, molecular size marker (λ DNA digested with *Pst*I).

Table 3. In order to assess the proportion of clones containing repetitive or unique sequences, the hybridization of labelled *V. faba* genomic DNA to the inserts was performed (Figure 6). Of 182 clones tested, 76 (42%) showed a strong or intermediate hybridization signal suggesting that they originated from highly or moderately repetitive genomic sequences. The remaining inserts (58%) contained unique or low-copy sequences showing no or a very weak hybridization signal. The results of dot-blot hybridization were confirmed by Southern hybridization of selected clones to digested field bean genomic DNA (not shown).

Isolation of clones containing microsatellite sequences

Screening of 3500 colonies by PCR using a combination of microsatellite [$(GA)_{11}$ or $(CAA)_7$] and plasmid-derived primers revealed 7 and 12 positive clones respectively. The colonies were retrieved from microtitre plates, and six of them were sequenced. With one exception, all inserts contained the respective microsatellite motifs (Table 4).

Table 3. Basic characteristics of chromosome-specific libraries

Library ID no.*	Chromosome type†	Starting chromosomes	Purity (%)	Number of colonies‡	Total library size§	Frequency of non-recombinants	Average insert length (bp)
Vf 1 EF 01	1 Std	1 000	96	2.9×10^4(3)	9.7×10^5	0.06	454
Vf 2 EF 01	2 F	1 000	94	4.5×10^4(3)	1.5×10^6	0.08	470
Vf 3 EF 01	3 F	1 000	ND**	2×10^4(2)	1×10^6	0.06	487
Vf 4 EF 01	4 E	1 000	96	4.6×10^4(3)	1.5×10^6	0.04	459
Vf 5 EF 01	5 E	1 000	ND**	1.2×10^4(2)	6×10^5	0.06	476
Vf 6 EF 01	6 Std	1 000	94	1.1×10^5(3)	3.7×10^6	0.04	424
Vf 1 JF 01	1 J	250	96	9×10^4(2)	4.5×10^6	0.05	310
Vf G 01	2 ng of genomic DNA		−	7×10^4(2)	3.5×10^6	0.05	433

*Library ID no. consists of abbreviation of species name (Vf), indication of chromosome and the karyotype used for sorting (e.g. 1 EF) and construction number. †The item shows whether the chromosome is wild-type (Std) or translocated [abbreviations of translocation types (E, F or J) are according to Schubert *et al.* (1986)]. See also Figure 1. ‡Number of colonies obtained after *E. coli* transformation using an aliquot of the ligation mix (the volume in μl is indicated in brackets). §Number of clones available after processing all amplified DNA from the corresponding sorted sample (estimated from previous item). **ND, chromosome purity not determined.

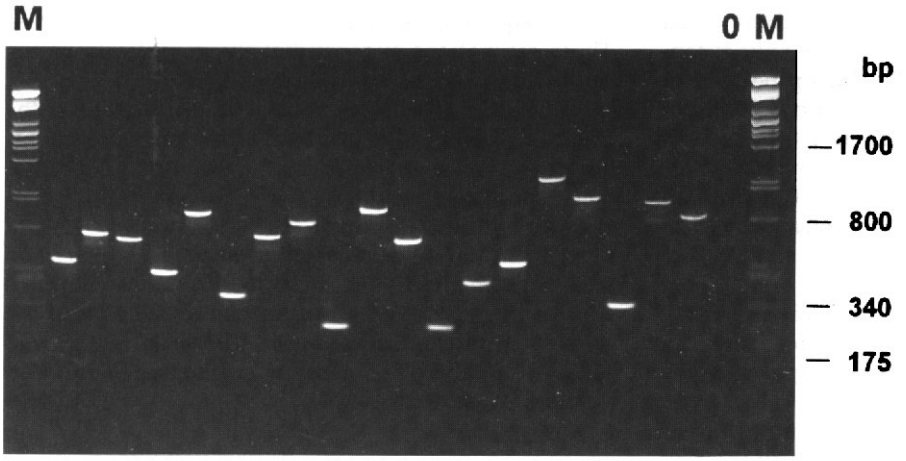


Figure 5. Size distribution of cloned fragments sampled from the library Vf 6 EF01. The inserts were PCR amplified using T3 and T7 primers and resolved on 2% agarose gel. M, molecular size marker (λ DNA digested with *PstI*); 0, no DNA (negative control).

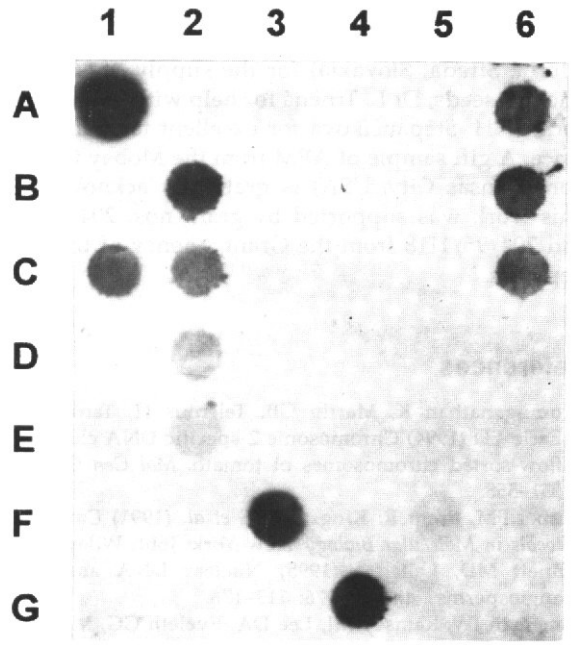


Figure 6. Dot-blot hybridization of 41 randomly selected clones from the library Vf 1 EF01 with labelled total field bean genomic DNA. Rows and columns of the dot-blot are indicated as A–G and 1–6 respectively. Cloned fragment of *V. faba* 45S rDNA [clone Ver17, Yakura & Tanifuji (1983)] as a control representing repetitive genomic sequence is at position A-1.

Discussion

The aim of this study was to prepare a set of chromosome-specific DNA libraries spanning the whole *V. faba* genome, which could serve as a basis for high-density molecular mapping. This was achieved by sorting and cloning of all six chromosome types from reconstructed karyotype EF. A library was also constructed from chromosome 1 of the karyotype JF, which has significantly reduced size in comparison with the wild-type chromosome 1 (Figure 1).

Table 4. Microsatellite sequences found in the clones selected from the library Vf 1 JF01

Clone	Microsatellite
JF1-AG2	$(AG)TC(AG)A(AG)_2AT(AG)_7$
JF1-AG3	$(GA)_5TA(GA)_4TA(GA)_{11}$
JF1-AAC1	$(AAC)_2N_{13}A_4(AAC)_4$
JFI-AAC2	$(AAC)_4A(AAC)_3TC(AAC)$
JF1-AAC3	$(AAC)_4TAATTC(AAC)$

The choice of the strategy for library construction depends on its application. Direct cloning permits generation of cosmid, phage or yeast artificial chromosome (YAC) libraries containing large fragments of chromosomal DNA, useful for gene isolation and assembly of contigs (Van Dilla & Deaven 1990, McCormick *et al.* 1993, Riess *et al.* 1994). Preparation of complex libraries by direct cloning requires relatively large amounts of starting material. In plants, this approach was used for library construction from a sample of 100 000 flow-sorted wheat chromosomes enriched for chromosome 4A (Wang *et al.* 1992). In this study, the DNA was cleaved by *Hpa*II to clone unmethylated low copy sequences preferentially. On the other hand, short-insert libraries may be used as a rich source of chromosome-specific molecular markers (Arumuganathan *et al.* 1994, Riquet *et al.* 1995). Arumuganthan *et al.* (1994) described the preparation of a library from 1000 flow-sorted tomato chromosomes number 2 after DOP-PCR amplification. The library was used for isolation of restriction fragment length polymorphism (RFLP) markers from this chromosome type.

In this work, we have chosen the same strategy. The requirement for a low number of sorted chromosomes allowed us to employ two-step sorting, which resulted in highly purified fractions. This was demonstrated by microscopic observation. Specific reamplification of three genes from DOP-PCR-amplified samples provided additional proof of their purity. Only chromosome

types known to contain respective sequences showed positive signals (Figure 4 & Table 2). The only exception was the amplification of USP, which produced an extra band in the sample of chromosome 2 EF. Since USP genes are located on chromosome 1 (Macas *et al.* 1993a, Fuchs & Schubert 1995), we presume that this result was caused by contamination of the chromosome 2 EF fraction by broken chromosome 1 EF (without the satellite), which is equal in size to the chromosome 2 EF.

The protocol for amplification and cloning of DOP-PCR products was modified extensively for increased efficiency. Our results suggest a limited capacity of the reaction to generate perfectly double-stranded DNA fragments. This could be a result of self-annealing of partially homologous fragments or caused by the lack of some of the reaction components. Thus, in contrast to the preparation of chromosome-painting probes in which single-stranded DNA does not represent a serious obstacle, the quality of DOP-PCR products intended for cloning should be preferred over the quantity, and a lower number of amplification cycles should be used. The inefficient cloning via the *Xho*I restriction site present on the primer 6-MW was presumably a result of the low ability of the enzyme to cleave close to the DNA termini (Jung *et al.* 1993). Replacement of the *Xho*I restriction site with the recognition sequences for endonucleases, which are known to work better near the ends of DNA fragments, led to a significant increase in cloning efficiency.

The amplified chromosomal DNA appeared as a smear of fragments 100–2500 bp long on agarose gel, and the size of most of the cloned fragments ranged between 100 and 800 bp. Similar size distributions were reported for DOP-PCR-amplified DNA from various species (Arumuganathan *et al.* 1994, Hoebee *et al.* 1994, Pich *et al.* 1994). Only about 40% of the analysed inserts contained repetitive genomic sequences. Taking into account the very high proportion of repetitive DNA in the field bean genome (more than 90%), this finding suggests preferential amplification of low-copy sequences. A similar observation was reported by other authors (Telenius *et al.* 1992, Arumuganathan *et al.* 1994). A high proportion of low-copy genomic sequences and relatively short inserts are advantageous features of DOP-PCR-generated libraries intended for isolation of molecular markers. Moreover, it is important that DOP-PCR products originate from multiple sites evenly distributed along the chromosomes, as was shown by *in situ* hybridization (Telenius *et al.* 1992, Hoebee *et al.* 1994).

Given the average insert size of 400 bp and 10^6 clones per library, the whole library contains approximately 400 Mbp of DNA, from which 240 Mbp (60%) are low-copy sequences. Considering the fact that an average field bean wild-type acrocentric chromosome contains approximately 2000 Mbp, and of this 90% comprises repetitive DNA, low-copy sequences represent 200 Mbp. Thus, the library contains approximately 1.2 equivalents of chromosomal low-copy DNA. These estimations suggest that the complexity of the libraries should be sufficiently high.

This report gives just the basic characteristics of the libraries. A more detailed evaluation of their composition, as well as their use for the isolation of chromosome-specific molecular markers, will be the subject of further studies. However, the isolation of several clones containing microsatellite sequences described here has already proved the suitability of the libraries for this purpose. We expect that the availability of the whole set of chromosome-specific libraries will greatly aid in the analysis and mapping of the large field bean genome.

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